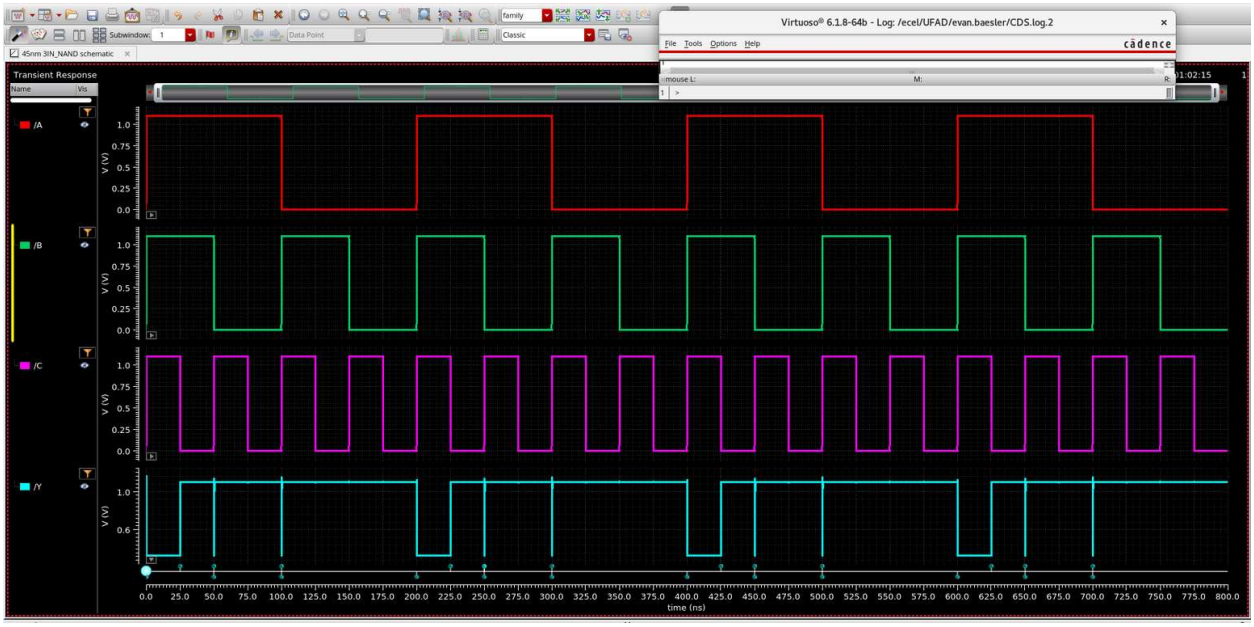
1. 3-INPUT NAND TRANSIENT (45nm)



2. 3-INPUT NAND (45nm)
   (exceptions on bounces, consistent otherwise.)
   Transient Rise & Fall (Output) for WP = WN = 1
   Rise: 9.685ps
   Fall: 14.42ps

   Transient Rise & Fall (Output) for WP = 1.25, WN = 1
   Rise: 9.699ps
   Fall: 14.99ps

   Transient Rise & Fall (Output) for WP = 1.5, WN = 1

   Rise: 9.547ps

   Fall: 15.23ps

   Transient Rise & Fall (Output) for WP = 1, WN = 1.25
   Rise: 9.875ps
   Fall: 13.66ps

   Transient Rise & Fall (Output) for WP = 1, WN = 1.5

   Rise: 10.12ps

   Fall: 13.12ps

   Same around: WP = 1, WN = 1.55

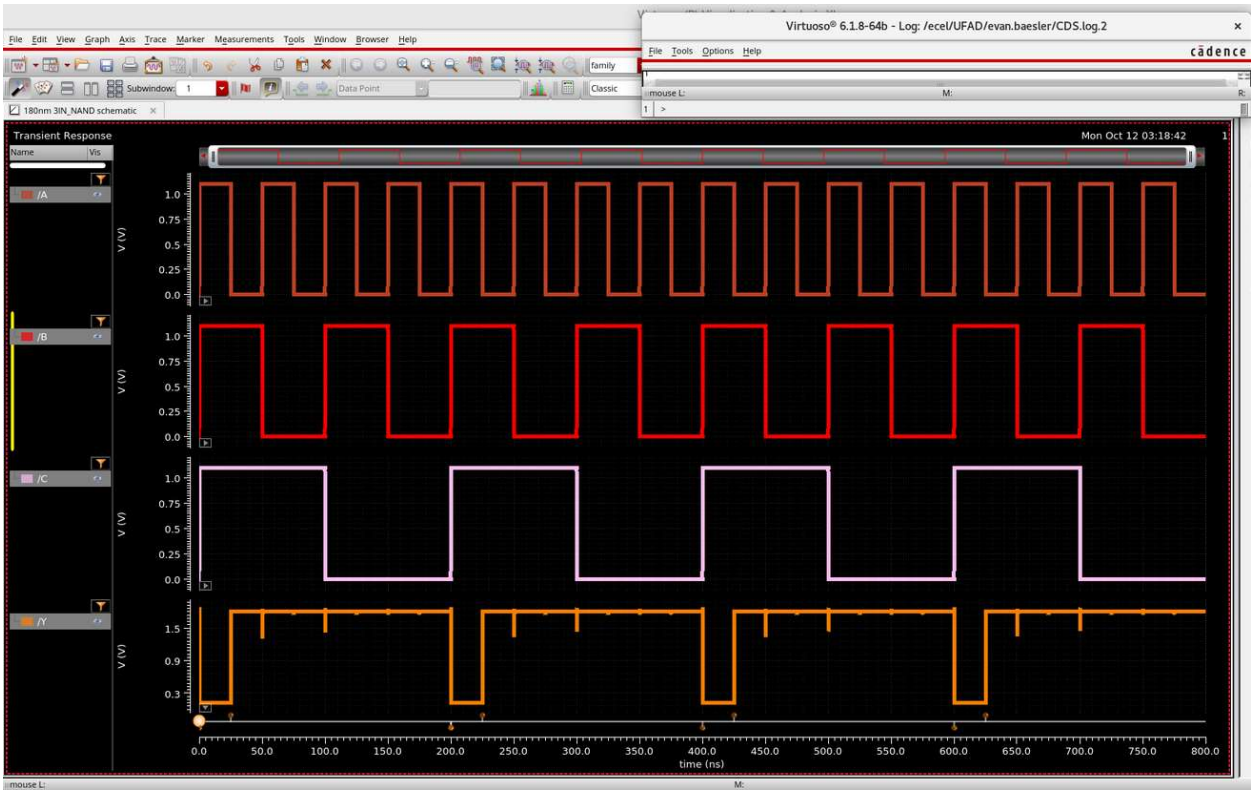
   Rise: 17.49ps

   Fall: 16.59ps

   17.49 – 16.59 = .9, just over 5%

   So, we get 186nm NMOS, 120nm PMOS

3. 3-INPUT NAND TRANSIENT (180nm)



4. 3-INPUT NAND (180nm)

   Transient Rise & Fall (Output) for WP = WN = 1
   Rise: 24.16ps
   Fall: 45.17ps

   Transient Rise & Fall (Output) for WP = 1, WN = 1.25
   Rise: 35.67ps
   Fall: 22.12ps

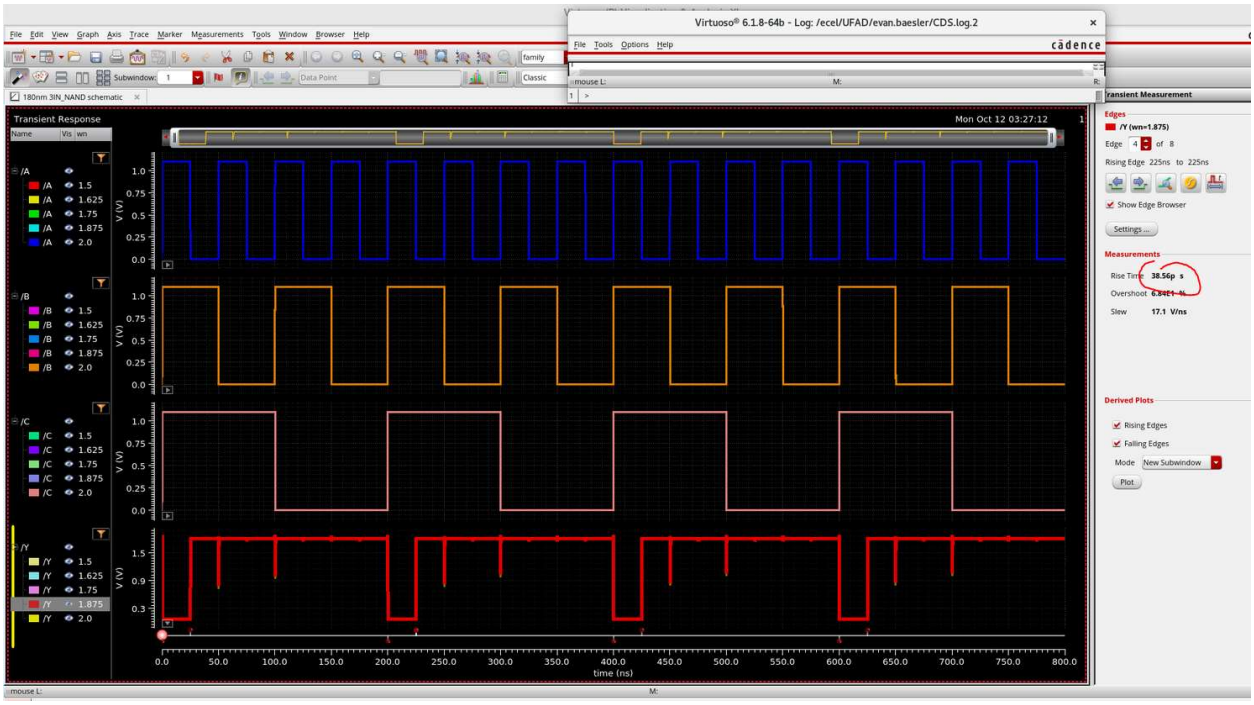
   Transient Rise & Fall (Output) for WP = 1, WN = 1.5
   Rise: 20.4ps
   Fall: 29.86ps

   Transient Rise & Fall (Output) for WP = 1.25, WN = 1
   Rise: 24.16ps
   Fall: 45.17ps

   Transient Rise & Fall (Output) for WP = 1.5, WN = 1
   Rise: 26.7ps
   Fall: 83.08ps

Closest found: PW = 1, WN = 1.875:

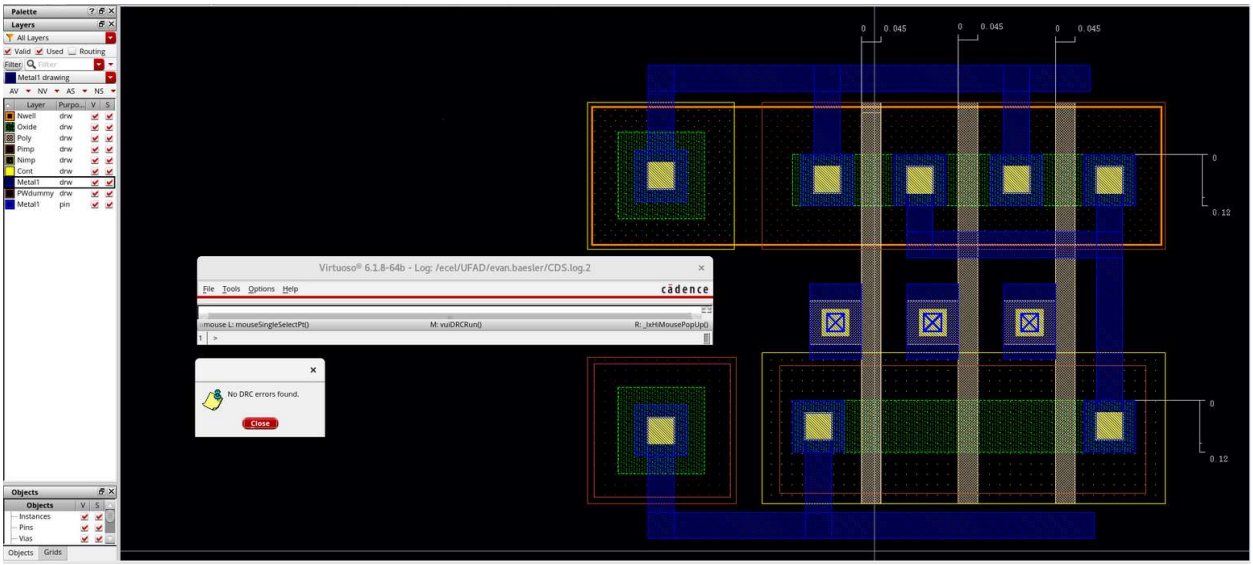
Rise: 38.56ps
Fall: 40.09ps
Diff: 40.09-38.56 = 1.53, within 5% of eachother.



5. Naturally, the 45nm technology node is significantly faster and has quicker rise and fall times, which were about 17ps avg rise/fall versus 39.5ps avg rise/fall, which means the smaller technology node is denser and over twice as fast as the previous nodes. This is likely due to the fact that smaller dimensions means smaller parasitic capacitance, which allows for faster charging and discharging.

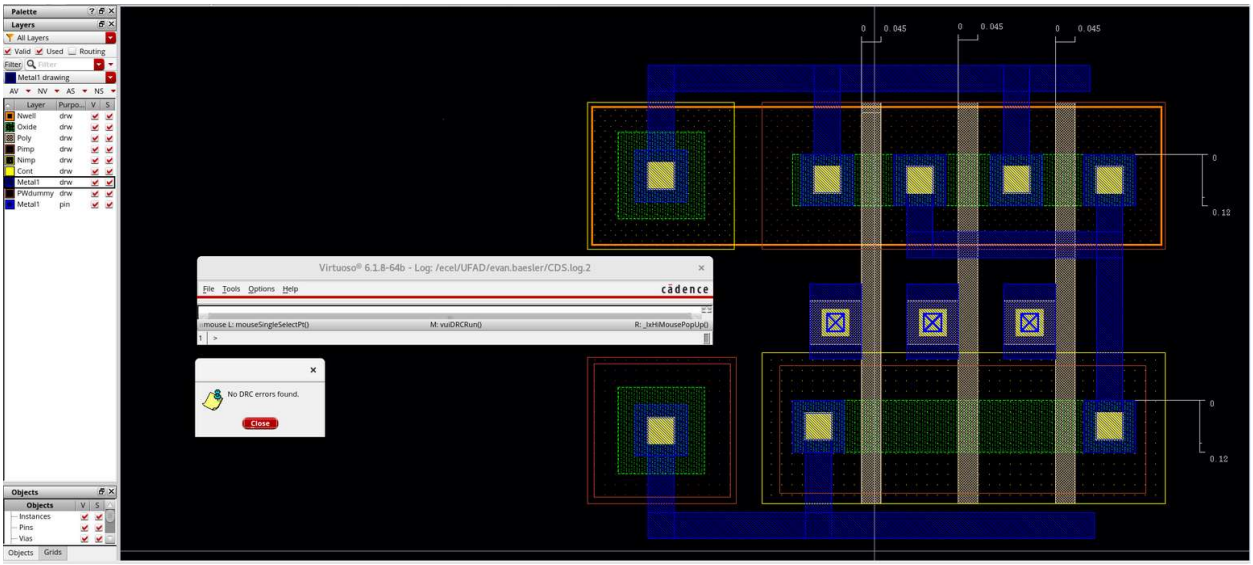
   To size these two transistors effectively, I tried to keep their differences within 5% of each other, which brought the sizes up considerably. Both transistors found a need for increasing the WN value, to bring the fall time down. I think that the sizing was different between the two transistors due to differing materials and nuance behind the parasitics.

6. Used 120nm transistor size standard:

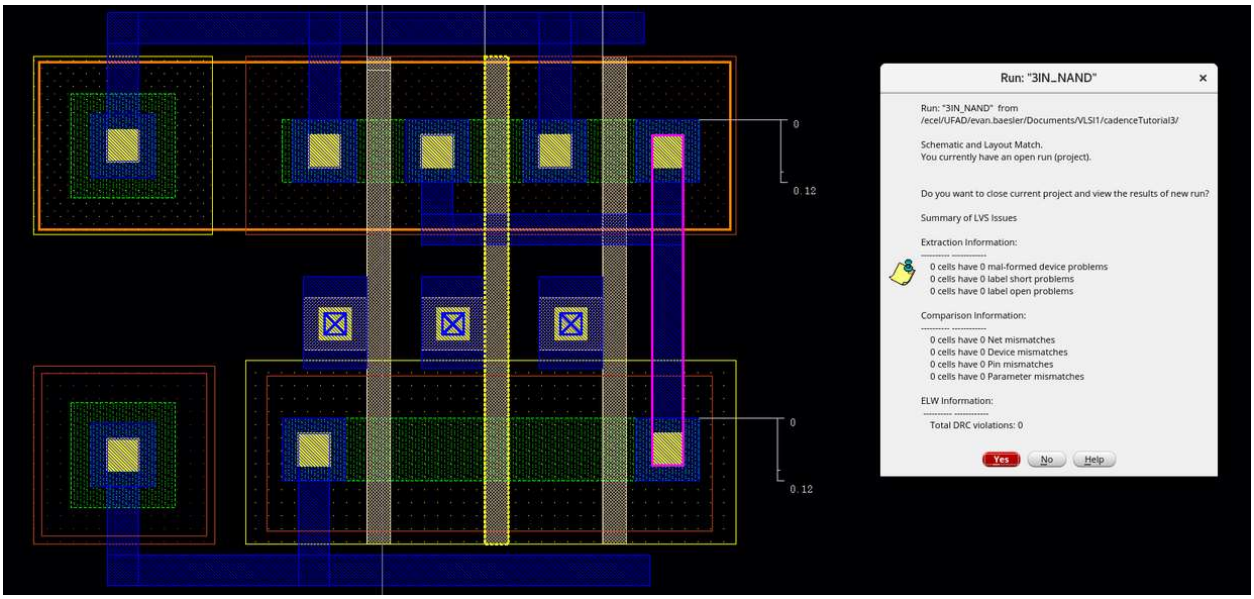


7. The 180nm NAND is expected to be larger than the 45nm NAND because its transistors will generally be larger in terms of width and length. Ideally, the ratio of area would be that the 180nm would be: $\frac{180nm^2}{45nm^2} = 16$ times larger than the 45nm NAND. Although, with the different NMOS sizing this might fluctuate slightly.

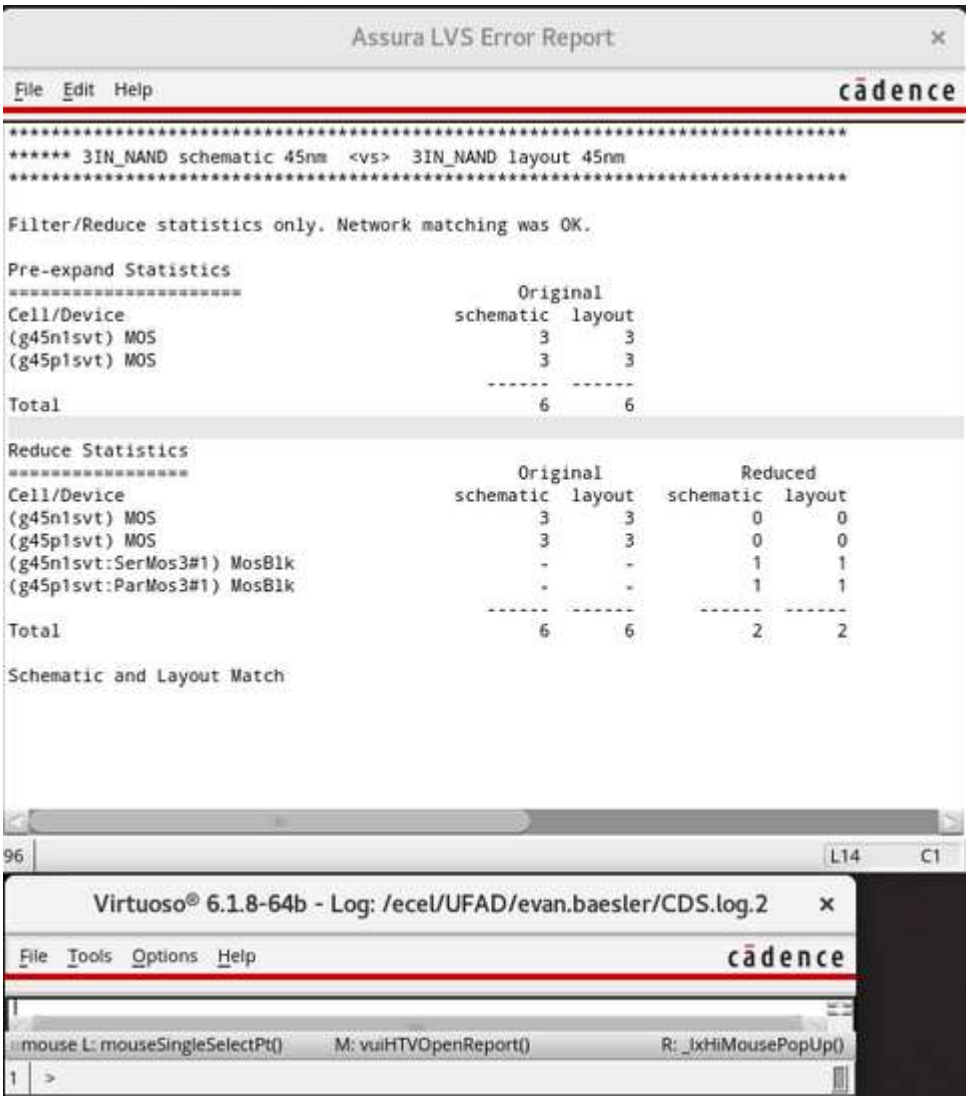
8. DRC:



9. LVS


0
0.12
0
0.12
Run: "3IN_NAND"
Run: "3IN_NAND" from
/ecel/UFAD/evan.baesler/Documents/VLSI1/cadenceTutorial3/
Schematic and Layout Match.
You currently have an open run (project).
Do you want to close current project and view the results of new run?
Summary of LVS Issues
Extraction Information:
0 cells have 0 mal-formed device problems
0 cells have 0 label short problems
0 cells have 0 label open problems
Comparison Information:
0 cells have 0 Net mismatches
0 cells have 0 Device mismatches
0 cells have 0 Pin mismatches
0 cells have 0 Parameter mismatches
ELW Information:
Total DRC violations: 0
Yes
No
Help

10. LVS Output:



Assura LVS Error Report

File Edit Help cadence

```
********************************************************************************
****** 3IN_NAND schematic 45nm  <vs>  3IN_NAND layout 45nm
********************************************************************************

Filter/Reduce statistics only. Network matching was OK.

Pre-expand Statistics
=====================                    Original
Cell/Device                           schematic  layout
(g45n1svt) MOS                            3       3
(g45p1svt) MOS                            3       3
                                        ------  ------
Total                                     6       6

Reduce Statistics
=================                        Original           Reduced
Cell/Device                           schematic  layout   schematic  layout
(g45n1svt) MOS                            3       3          0       0
(g45p1svt) MOS                            3       3          0       0
(g45n1svt:SerMos3#1) MosBlk               -       -          1       1
(g45p1svt:ParMos3#1) MosBlk               -       -          1       1
                                        ------  ------     ------  ------
Total                                     6       6          2       2

Schematic and Layout Match
```

96 L14 C1

Virtuoso® 6.1.8-64b - Log: /ecel/UFAD/evan.baesler/CDS.log.2

File Tools Options Help cadence

mouse L: mouseSingleSelectPt() M: vuiHTVOpenReport() R: _lxHiMousePopUp()

1 >